

GSP (Generalised Sequential Patterns)

SID	Sequence
1	$\langle \{a, b\}, \{c\}, \{f, g\}, \{g\}, \{e\} \rangle$
2	$\langle \{a, d\}, \{c\}, \{b\}, \{a, b, e, f\} \rangle$
3	$\langle \{a\}, \{b\}, \{f, g\}, \{e\} \rangle$
4	$\langle \{b\}, \{f, g\} \rangle$

- Min sup = 2

1- length Seq	Support
{a}	3
{b}	4
{c}	2
{d}	1
{e}	3
{f}	4
{g}	3

Candidates are generated in two steps:

1. **Join Phase.** We generate candidate sequences by joining L_{k-1} with L_{k-1} . A sequence s_1 joins with s_2 if the subsequence obtained by dropping the first item of s_1 is the same as the subsequence obtained by dropping the last item of s_2 . The candidate sequence generated by joining s_1 with s_2 is the sequence s_1 extended with the last item in s_2 . The added item becomes a separate element if it was a separate element in s_2 , and part of the last element of s_1 otherwise. When joining L_1 with L_1 , we need to add the item in s_2 both as part of an itemset and as a separate element, since both $\langle (x) (y) \rangle$ and $\langle (x y) \rangle$ give the same sequence $\langle (y) \rangle$ upon deleting the first item. (Observe that s_1 and s_2 are contiguous subsequences of the new candidate sequence.)
2. **Prune Phase.** We delete candidate sequences that have a contiguous $(k-1)$ -subsequence whose support count is less than the minimum support. If there is no max-gap constraint, we also delete candidate sequences that have any subsequence without minimum support.

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- $\{ab\}$ not equal to $\{ba\}$
- $\{(a,b)\}$ equal to $\{(b,a)\}$

1- length Seq	Support	2-len Seq (a)	
$\{a\}$	3	$\{ab\}$	2
$\{b\}$	4	$\{ac\}$	2
$\{c\}$	2	$\{ae\}$	3
$\{e\}$	3	$\{af\}$	3
$\{f\}$	4	$\{ag\}$	2
$\{g\}$	3	$\{(a,b)\}$	2
		$\{(a,c)\}$	0
		$\{(a,e)\}$	1
		$\{(a,f)\}$	1
		$\{(a,g)\}$	0

1- length Seq	support	2-len Seq (a)	2-len Seq (b)	2-len Seq (c)	2-len Seq (e)	2-len Seq (f)	2-len Seq (g)
{a}	3	{ab}	{ba}	{ca}	{ea}	{fa}	{ga}
{b}	4	{ac}	{bc}	{cb}	{eb}	{fb}	{gb}
{c}	2	{ae}	{be}	{ce}	{ec}	{fc}	{gc}
{e}	3	{af}	{bf}	{cf}	{ef}	{fe}	{ge}
{f}	4	{ag}	{bg}	{cg}	{eg}	{fg}	{gf}
{g}	3	{(a,b)}	{(b,c)}	{(c,e)}	{(e,f)}	{(e,g)}	
		{(a,c)}	{(b,e)}	{(c,f)}	{(e,g)}		
		{(a,e)}	{(b,f)}	{(c,g)}			
		{(a,f)}	{(b,g)}				
		{(a,g)}					

[illegible]

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2-len Seq

{ab}	2
{ac}	2
{ae}	3
{af}	3
{ag}	2
{(a,b)}	2
{be}	3
{bf}	4
{bg}	3
{ce}	2
{cf}	2
{fe}	2
{(f,g)}	3
{ge}	2

3-len Seq

{abe}	{(ab)f}
{abf}	{(ab)g}
{abg}	{bfe}
{ace}	{b(f,g)}
{acf}	{bge}
{afe}	{cfe}
{a(f,g)}	{c(f,g)}
{age}	{(f,g)e}
{(ab)e}	

- {ab} {ae} {be}
- {cf} {cg} {(f,g)}

-
- ~~{ab}~~ ~~{ac}~~
 - ~~{ab}~~ ~~{be}~~ → {abe}
 - ~~{(a,b)}~~ ~~{be}~~ → {(a,b)e}
 - ~~{bf}~~ ~~{(f,g)}~~ → { b(f,g) }

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3-len Seq			
{abe}	2	{(ab)e}	1
{abf}	2	{(ab)f}	1
{abg}	1	{(ab)g}	1
{ace}	2	{bfe}	2
{acf}	2	{b(f,g)}	3
{afe}	2	{bge}	2
{a(f,g)}	2	{cfe}	1
{age}	2	{(f,g)e}	2

3-len Seq	
{abe}	2
{abf}	2
{ace}	2
{acf}	2
{afe}	2
{a(f,g)}	2
{age}	2
{bfe}	2
{b(f,g)}	3
{bge}	2
{(f,g)e}	2

4-len Seq	
{abfe}	1
{ab(f,g)}	1
{a(f,g)e}	2
{b(f,g)e}	2

- {abf} {bfe} $\longrightarrow$ {abfe}
- {abf} {bge}

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1- length Seq		2-len Seq		3-len Seq		4-len Seq	
{a}	3	{ab}	2	{abe}	2	{abfe}	2
{b}	4	{ac}	2	{abf}	2	{a(f,g)e}	2
{c}	2	{ae}	3	{ace}	2	{b(f,g)e}	2
{e}	3	{af}	3	{acf}	2		
{f}	4	{ag}	2	{afe}	2		
		{(a,b)}	2	{a(f,g)}	2		
{g}	3	{be}	3	{age}	2		
		{bf}	4	{bfe}	2		
		{bg}	3	{b(f,g)}	3		
		{ce}	2	{bge}	2		
		{cf}	2	{(f,g)e}	2		
		{fe}	2				
		{(f,g)}	3				
		{ge}	2				